



Xin-Yu Xiao

AI for Science, NLP Applications, LLMs
University of Chinese Academy of Sciences
National Astronomical Observatories, CAS

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Github Homepage

LinkedIn Profile

EDUCATION

- University of Chinese Academy of Sciences** *Sep. 2024 – Present*
M.S. in Astronomical Techniques and Methods GPA: 3.63/4.00
- Qilu University of Technology (Shandong Academy of Sciences)** *Sep. 2020 – Jun. 2024*
B.S. in Computer Science and Technology GPA: 4.02/5.00

RESEARCH EXPERIENCE



The Future Laboratory, Tsinghua University ↗

Sep. 2024 – Present

Research Intern Advisor: Qianchen Xia, Assistant Professor

Beijing, China

- Built a multi-agent data generation framework that integrates instruction generation and iterative refinement to support training *Lunar Twins*, the first LLM specialized for lunar exploration.
- Developed *Lunar-Bench* to evaluate task-oriented LLM reasoning in lunar contexts, and built a UE5-based simulation environment for representative surface operations and human-AI interaction.
- Proposed *Lunar-KAG*, a knowledge-augmented generation framework combining hypergraph knowledge representation with dual-path retrieval, which enhances scientific knowledge grounding while reducing redundant reasoning.
- Engineered *Lunar-R1* using enhanced GRPO reinforcement learning for efficient reasoning in lunar missions. It optimizes the accuracy-efficiency trade-off and is deployed on a resource-constrained lunar rover platform.



Southern University of Science and Technology (SUSTech) ↗

Apr. 2024 – Aug. 2024

Visiting Student Advisor: Dikun Yang, Associate Professor

Shenzhen, China

- Participated to a National Key R & D Program project on transient electromagnetic sensing and imaging, and led an twelve-week field survey at the Daqing Oilfield for large-scale electromagnetic data acquisition.
- Designed and deployed an AWS serverless architecture (Lambda, S3, EC2) to automate data pipelines, enabling real-time synchronization and centralized management for 100+ remote acquisition units.
- Implemented time-series signal preprocessing and electromagnetic response inversion workflows for field measurements, supporting subsurface imaging and electromagnetic characterization of shale gas reservoirs.



National Astronomical Observatories, Chinese Academy of Sciences ↗

Jun. 2023 – Jun. 2024

Research Intern Advisor: Qiao Wang, Associate Professor

Beijing, China

- Contributed to the Virtual Universe Modeling and Visualization project, applying high-resolution N-body and cosmological hydrodynamic simulations to visualize cosmic structure formation.
- Designed data-processing workflows to extract galaxies, dark matter, and large-scale structures from cosmological simulations, and implemented visualization pipelines for interactive rendering.
- Implemented GPU-accelerated rendering and delivered an immersive presentation of cosmological simulations, which were exhibited at the National Museum of China and the Art Museum of the China Academy of Art.

SELECTED PUBLICATIONS (*CORRESPONDING)

- [1] Xin-Yu Xiao, Yalei Liu, Xiangyu Liu, Zengrui Li, Erwei Yin, Qianchen Xia*. Lunar Twins: We Choose to Go to the Moon with Large Language Models. *ACL 2025, Accepted & Published*. [PDF]
- [2] Xin-Yu Xiao, Ye Tian, Erwei Yin, Zhixian He, Shiqi Wang, Qianchen Xia*. Lunar-Bench: Evaluating Task-Oriented Reasoning of LLMs in Lunar Exploration Scenarios. *ACL 2026, Accepted & Published*. [PDF]
- [3] Xin-Yu Xiao, Zhixian He, Ye Tian, Shiqi Wang, Qianchen Xia*. Lunar-R1: Towards Efficient Reasoning of LLMs for Lunar Exploration via Reinforcement Learning. *EMNLP 2026, Accepted*. [PDF]
- [4] Xin-Yu Xiao, Zhixian He, Qilin Chen, Qianchen Xia*. Lunar-KAG: Knowledge Augmented Generation for Scientific Discovery in Lunar Exploration. *Under Review*. [PDF]
- [5] Xin-Yu Xiao*, Hanyu Yang, Jun Zhu. Hypergraph Label-Element Neural Networks for Low-Resource Chinese Legal Event Detection. *PRICAI 2026, Accepted*.

HONORS AND SKILLS

Honors: Academic Scholarship, UCAS (2024–2027); National Scholarship (2023); Jinan Quancheng Scholarship (2023); National Encouragement Scholarship (2022); Academic Scholarship, Qilu University of Technology (2021–2023).

Programming: Python, C++, SQL, R, MATLAB **Tools:** GIS, ENVI, Git, Origin, Photoshop, Illustrator, UE, Blender

Languages: Chinese (Native), English (Basic), German (Basic).